Problem Set 2

AI 501: Mathematics for Artificial Intelligence

Fall 2026

Problem 1. Let

$$x = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \qquad y = \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix}.$$

Calculate:

(a) $x^T y$;

(b) $|x|$ and $|y|$;

(c) the distance between x and y;

(d) the cosine of the angle between x and y.

Based only on the sign of $x^T y$, before calculating the angle, would you expect the angle to be acute, right, or obtuse?

Problem 2. Let

$$u = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \qquad v = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}.$$

Find a nonzero vector

$$w = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

such that w is orthogonal to both u and v.

Is your answer unique? Briefly explain.

Problem 3. Let x and y be nonzero vectors.

For each statement, determine whether it is *always true*, *sometimes true*, or *never true*. Give a counterexample when appropriate.

(a) If $x^T y = 0$, then x and y have the same length.

(b) If $|x| = |y|$, then $x^T y = 0$.

(c) If $x^T y > 0$, then the angle between x and y is acute.

(d) If $x^T y < 0$, then the angle between x and y is obtuse.

(e) If x and y are orthogonal, then $|x + y|^2 = |x|^2 + |y|^2$.

Problem 4. Let u and v be orthogonal vectors, and suppose

$$\|u\| = 3, \qquad \|v\| = 4.$$

Without finding the actual entries of u and v, calculate:

(a) $\|u + v\|$;

(b) $\|2u - v\|$;

(c) $\|3u + 4v\|$.

Problem 5. A vector x is replaced by

$$\hat{x} = \frac{x}{\|x\|}.$$

Let

$$x = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \qquad y = \begin{pmatrix} -4 \\ 3 \end{pmatrix}.$$

(a) Find $\hat{x}$ and $\hat{y}$.

(b) What is $\hat{x}^T \hat{y}$?

(c) Compare $x^T y$ with $\hat{x}^T \hat{y}$.

(d) What property of x and y is preserved by normalization?

Problem 6. Let

$$x = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \qquad y = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}, \qquad z = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}.$$

Without calculating any angles:

(a) Which pair has angle 0?

(b) Which pair has an acute angle?

(c) Which pair has an obtuse angle?

Verify your answers using inner products.

Problem 7. Two students are discussing the distance between vectors.

Student A says:

"If two vectors point in exactly the same direction, their distance must be zero."

Student B says:

> "If two vectors are orthogonal, their distance must be the same regardless of their lengths."

Determine whether each student is correct. Give a simple counterexample for every incorrect statement.

Problem 8. Let $x = (x_1, \ldots, x_n)^T$ and let

$$\mathbf{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}.$$

The mean of the entries of x is

$$\mu = \frac{1}{n}\mathbf{1}^T x.$$

Define the centered vector

$$x_c = x - \mu\mathbf{1}.$$

(a) Show that the entries of x_c sum to zero.

(b) Show that

$$\mathbf{1}^T x_c = 0.$$

(c) What does this tell you geometrically about x_c and $\mathbf{1}$?

Problem 9. Consider the two datasets

$$x = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \qquad y = \begin{pmatrix} 10 \\ 20 \\ 30 \\ 40 \end{pmatrix}.$$

(a) Find their means and standard deviations.

(b) Find the standardized versions of x and y.

(c) Calculate the correlation coefficient between x and y.

Now replace y by

$$z = \begin{pmatrix} 40 \\ 30 \\ 20 \\ 10 \end{pmatrix}.$$

Without performing the full calculation, predict the sign of the correlation between x and z. Then verify your prediction.

Problem 10. Let

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad v_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

(a) Express v_3 as a linear combination of v_1 and v_2.

(b) Is v_3 necessary to span the same set of vectors as v_1, v_2, v_3?

(c) Describe $\operatorname{span} v_1, v_2, v_3$ geometrically.

(d) Are v_1, v_2, v_3 linearly independent?

Problem 11. Let

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad v_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Determine whether

$$b = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}$$

belongs to $\operatorname{span} v_1, v_2, v_3$.

If it does, find scalars c_1, c_2, c_3 such that

$$b = c_1 v_1 + c_2 v_2 + c_3 v_3.$$

Problem 12. Let

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad v_4 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Without performing Gaussian elimination, determine whether the vectors are linearly independent.

Then answer:

(a) Do they span $\mathbb{R}^4$?

(b) Do they form a basis of $\mathbb{R}^4$?

Give a short justification.

Problem 13. For each of the following sets, determine whether it is a basis for $\mathbb{R}^3$.

(a)

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

(b)
$$\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}$$

(c)
$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

For each set that is not a basis, identify the reason: failure of spanning, failure of linear independence, or both.

Problem 14. Let

$$u = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}.$$

(a) Show that u and v are orthogonal.

(b) Normalize both vectors to obtain q_1 and q_2.

(c) Show that q_1, q_2 is orthonormal.

(d) Write

$$x = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$$

as a linear combination of q_1 and q_2.

Problem 15. A teacher model produces the following probability distribution over four classes:

$$p = \begin{pmatrix} 0.10 \\ 0.60 \\ 0.20 \\ 0.10 \end{pmatrix},$$

while a student model produces

$$\hat{p} = \begin{pmatrix} 0.25 \\ 0.55 \\ 0.05 \\ 0.15 \end{pmatrix}.$$

Let the error vector be

$$e = \hat{p} - p.$$

(a) Compute the error vector e.

(b) Compute the Chebyshev norm $\|e\|_\infty$.

(c) Which output label has deviated the most from the teacher's prediction?

(d) In one sentence, explain why the Chebyshev norm is useful for this particular comparison.

Problem 16. Let u, v, w be nonzero vectors.

For each statement, determine whether it is true or false. If false, give a counterexample.

(a) Three mutually orthogonal nonzero vectors are linearly independent.

(b) Three linearly independent vectors in $\mathbb{R}^2$ can exist.

(c) Two orthonormal vectors must be linearly independent.

(d) Every basis is an orthonormal basis.

(e) Every orthonormal set of vectors is linearly independent.

(f) Four mutually orthogonal nonzero vectors can exist in $\mathbb{R}^3$.